A sample chapter

# 100 Puzzles

*Stories to Solve with Mathematics, Logic and Science*

The award book of Math Club & Competitions.

This sample holds the Prologue, where the journey begins, and Puzzle 8, The Swinging Island, with its hint and full solution. Try the puzzle before you turn to the answers.

MATH CLUB & COMPETITIONS



PROLOGUE

# The Blank Places

Snow had been falling on Highmere for three days when Master Orla called her four apprentices to the Hall of Maps. The Great Map filled the whole north wall: mountains in brown ink, rivers in blue, roads in fine red lines. Copies of it hung in every inn and ferry house in the land. Travellers planned their journeys by it, healers found their way to the sick by it, and merchants sent their wagons along its roads. But here and there the roads stopped short at an empty square of parchment. Mira had counted them once. There were a hundred.

"Every blank is a real place," said Orla, tapping one with her cane. "A harbour, a market, a village on a cliff. We know their names from travellers' tales, and nothing more. So the roads stop short of them. Wagons turn back, letters go astray, and the people there wait for help that does not know the way."

She looked at each of them in turn. Mira, the mapmaker's apprentice, had ink on her fingers. Tavi, the tinker's apprentice, was taking apart a lock that had not asked to be taken apart. Suri, from the apothecary, was writing in her notebook. Bao, the ferry runner, was eating somebody else's breakfast.

"You know the rule of this hall," said Orla. "Nothing goes on the Great Map that we have not seen with our own eyes and checked with our own heads. A guess on a map is worse than a blank, because someone will trust it."

"So we go and look," said Mira.

"You go and *help*," said Orla. "Every blank place has its trouble: a gate that will not open, a light that will not burn, a sum that nobody there can make come right. Strangers who walk past a trouble are shown the door. Strangers who stop and mend it are shown the roads. Mend it, truly and carefully, and the place will show you its paths, the way in and the way out. Then it can go on the map, and the roads can reach it at last."

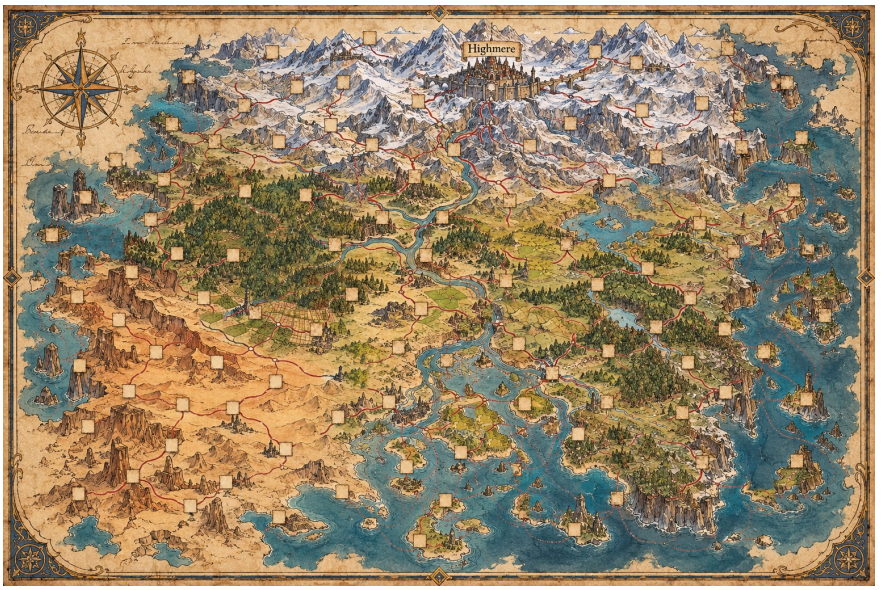
"All hundred?" said Bao, with his mouth full.

"By next midwinter, when you meet again at the mountain gate. Not alone, and not all together: you will travel by three roads and change companions as the seasons change. Mira carries the blank sheets and inks each place. Whoever is on another road sends their sketches to her."

A raven on the windowsill shook out her feathers, one of them silver, and counted the apprentices out loud. "One, two, three, four." Her name was Quill, and the letters would be hers to carry.

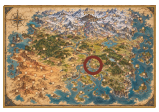
Orla's voice grew quieter. "One more thing. Long ago, someone else went looking for the blank places. If you ever find signs that you are not the first, write them down."

That night the four of them set out under falling snow: three lanterns going three different ways down the mountain, and one raven flying between them. Behind them, a hundred empty squares waited to be filled.




*The Great Map of Highmere, with its hundred blank places.*





PUZZLE 8

# The Swinging Island

Snow was coming. Suri and Bao felt it in the wind as the floating monastery of Skytether creaked on its chain, high above a white valley of farms and frozen streams. The whole island was moored by one iron chain, fixed under the bell tower, and the winter wind pushed it slowly round that chain like a weathervane.

Sister Lumen, who kept the monastery's stores, spread an old survey map of the valley across the table. "Six villages are waiting for their winter baskets," she said. "Tonight the wind will turn us once all the way round before the storm. We lower baskets from the landing gate, and the gate goes wherever the island takes it. Nobody has ever managed to steer an island, though the novices keep trying."

The map was ruled in equal squares. The spot below the bell tower was marked at 8 squares across and 8 up from the map's corner, and the spot below the gate at 13 across and 13 up.

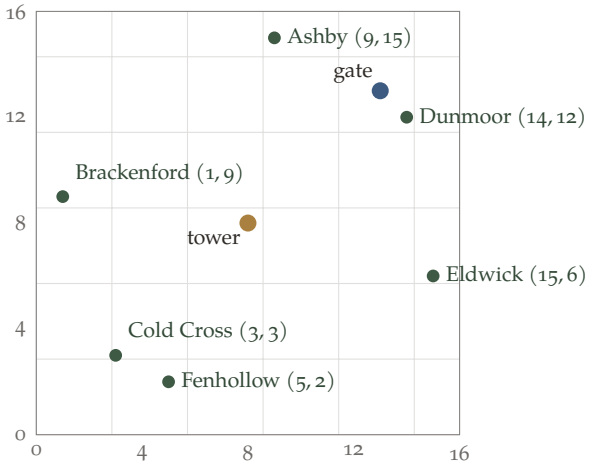
Suri traced a circle with her finger. "Anything that turns about one point keeps the same distance from it, like the handle of a door swinging round its hinges."

"Here is everything we may do," said Sister Lumen.

1. Lower a basket whenever the gate is exactly above a village.
2. The rope hangs straight down; a basket cannot be swung sideways.
3. The island turns one full circle, slowly, and cannot be steered, stopped or moved off its chain.

"A basket lowered anywhere else is lost in the snow."

Bao had already packed one for Dunmoor. "It's practically next to the gate!" Above them, the bell swayed, and the chain creaked again.




## WHAT YOU NEED TO KNOW

- Positions are counted in squares across and up from the map's corner. Villages, the tower spot and the gate spot all sit where grid lines cross.
- If one spot is $a$ squares across and $b$ squares up or down from another, the distance between them, squared, is $a^2 + b^2$ (Pythagoras).

## CAN YOU HELP SURI AND BAO?

**1.** Which villages will get their baskets? Is Bao right about Dunmoor?

**2.** Sister Lumen wants to build a new store hut at a crossing of grid lines that the gate will pass over. Find every such crossing, and explain why there are no others.

**3.** Two of the villages that get baskets are exactly a quarter turn apart around the bell tower. Which two? Explain how you can tell from the grid, without measuring an angle.

## BEFORE YOU READ ON

### HINTS

The gate always stays the same distance from the bell tower. Work with distance squared: for the gate it is $5^2 + 5^2$. Which other pairs of squares add up to the same number?

*Have you tried all three questions? The solution is overleaf.*

SOLUTION 8

# The Swinging Island

*Tags:* Circles; Pythagoras; Sums of squares; Quarter turns

## 1. The gate's path

The island turns about the spot below the bell tower, $T = (8, 8)$, so the gate always stays the same distance from $T$: it travels round a circle with $T$ at its centre. The gate starts 5 squares across and 5 squares up from $T$, so its distance squared is $5^2 + 5^2 = 50$.

A village gets a basket exactly when its distance from $T$, squared, is also 50. Working with distance squared avoids square roots, and two distances are equal exactly when their squares are.

## 2. The six villages

| Village | Across from $T$ | Up from $T$ | Distance squared | Basket? |
|---|---|---|---|---|
| Ashby | $+1$ | $+7$ | $1 + 49 = 50$ | yes |
| Brackenford | $-7$ | $+1$ | $49 + 1 = 50$ | yes |
| Cold Cross | $-5$ | $-5$ | $25 + 25 = 50$ | yes |
| Dunmoor | $+6$ | $+4$ | $36 + 16 = 52$ | no |
| Eldwick | $+7$ | $-2$ | $49 + 4 = 53$ | no |
| Fenhollow | $-3$ | $-6$ | $9 + 36 = 45$ | no |

Dunmoor is only a little too far out: the gate passes about a seventh of a square inside it. But the rope hangs straight down, so "nearly" is not enough.

> Ashby, Brackenford and Cold Cross get their baskets.
> Bao is wrong: Dunmoor is just outside the gate's circle.

## 3. Every crossing on the circle

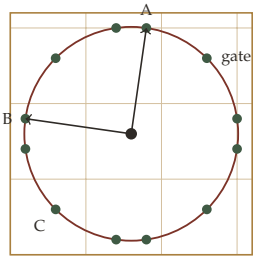
A crossing $a$ squares across and $b$ squares up or down from $T$ is on the circle exactly when $a^2 + b^2 = 50$. Each of $a^2$ and $b^2$ is a square number no bigger than 50: 0, 1, 4, 9, 16, 25, 36 or 49. Subtract each from 50:

$$50,\ 49,\ 46,\ 41,\ 34,\ 25,\ 14,\ 1.$$

Only 49, 25 and 1 are square numbers, so the only ways are $1 + 49$, $25 + 25$ and $49 + 1$. Each step can go either way (left or right, up or down), which gives

$$(\pm 1, \pm 7), \qquad (\pm 5, \pm 5), \qquad (\pm 7, \pm 1)$$

from $T$: four crossings each, twelve in all, and they all fit on the map. On the map they are $(9, 15)$, $(7, 15)$, $(9, 1)$, $(7, 1)$, $(1, 9)$, $(1, 7)$, $(15, 9)$, $(15, 7)$, $(3, 3)$, $(3, 13)$, $(13, 3)$ and $(13, 13)$.




Twelve crossings, shown as dots in the drawing. One of them, $(13, 13)$, is where the gate starts.

### 4. A quarter turn without angles

Turn the map a quarter turn anticlockwise about $T$. A step "1 right" becomes "1 up", and a step "7 up" becomes "7 left". So the journey from $T$ to Ashby, 1 right and 7 up, becomes 7 left and 1 up: exactly the journey from $T$ to Brackenford. The two arrows in the drawing are at right angles.

The same trick splits the twelve crossings into three groups of four, each turned into the next by a quarter turn. Written as steps from $T$:

$$(5, 5) \to (-5, 5) \to (-5, -5) \to (5, -5),$$
$$(1, 7) \to (-7, 1) \to (-1, -7) \to (7, -1),$$
$$(7, 1) \to (-1, 7) \to (-7, -1) \to (1, -7).$$

The last two groups are different: $(1, 7)$ and $(7, 1)$ are mirror images, not quarter turns of each other. Cold Cross, at $(-5, -5)$, is in the first group; Ashby and Brackenford are in the second, so Cold Cross is not a quarter turn from either of them.

Ashby and Brackenford: a quarter turn changes the step 1 right and 7 up into 7 left and 1 up.

On the map *Sister Lumen let Suri and Bao copy her survey map of Skytether's valley, and Suri added the circle the gate sweeps over the valley. Quill carried both to Mira.*